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Saveetha Institute of Medical And Technical Sciences
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AI-Based DDoS Detection & Automatic Prevention

CSA0707 – Computer Networks

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Introduction

- Distributed Denial-of-Service (DDoS) attacks overwhelm a network or service with excessive traffic.
- Such attacks can cause slow performance, service unavailability, and resource exhaustion.
- Traditional threshold-based detection may struggle with changing traffic patterns.
- Artificial Intelligence can identify suspicious patterns from network traffic characteristics.
- The proposed system combines network monitoring, Machine Learning, and automatic prevention.

Objective

- Develop a software-based system for real-time DDoS detection.
- Simulate normal and suspicious network traffic in a controlled local environment.
- Extract useful network features from observed traffic.
- Train a Machine Learning model to distinguish Normal and DDoS traffic.
- Automatically respond to detected malicious traffic.

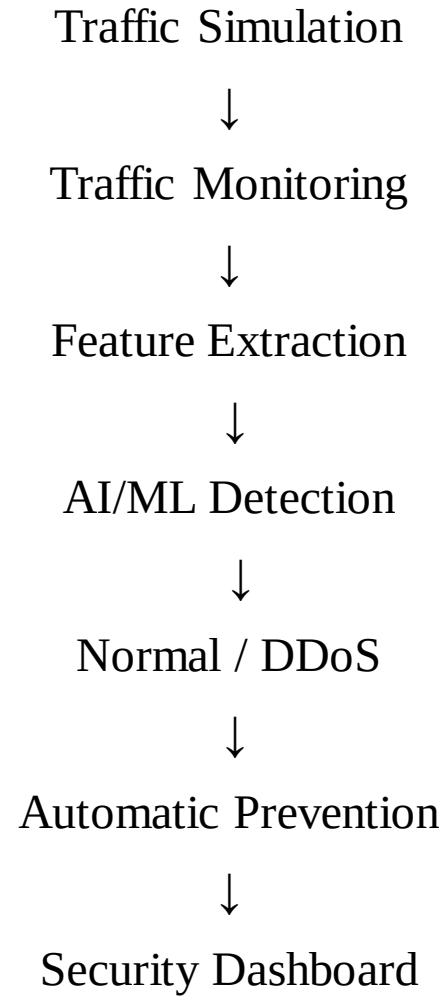
Problem Statement

- DDoS attacks can generate extremely high traffic within a short period.
- Sudden traffic spikes can consume server, network, and application resources.
- Static traffic thresholds may produce false positives or false negatives.
- Manual attack identification and response can introduce significant delay.
- Network administrators need continuous visibility into traffic behavior.
- Identifying suspicious source devices is difficult during high-volume traffic.

Proposed Methodology

- Generate controlled normal and DDoS-like traffic in a local simulation environment.
- Monitor incoming traffic and collect relevant network statistics.
- Extract features such as packet rate, request rate, byte rate, and source count.
- Feed the extracted features into a trained Random Forest classifier.
- Classify the observed traffic as Normal or DDoS.
- Generate a confidence score for the AI prediction.

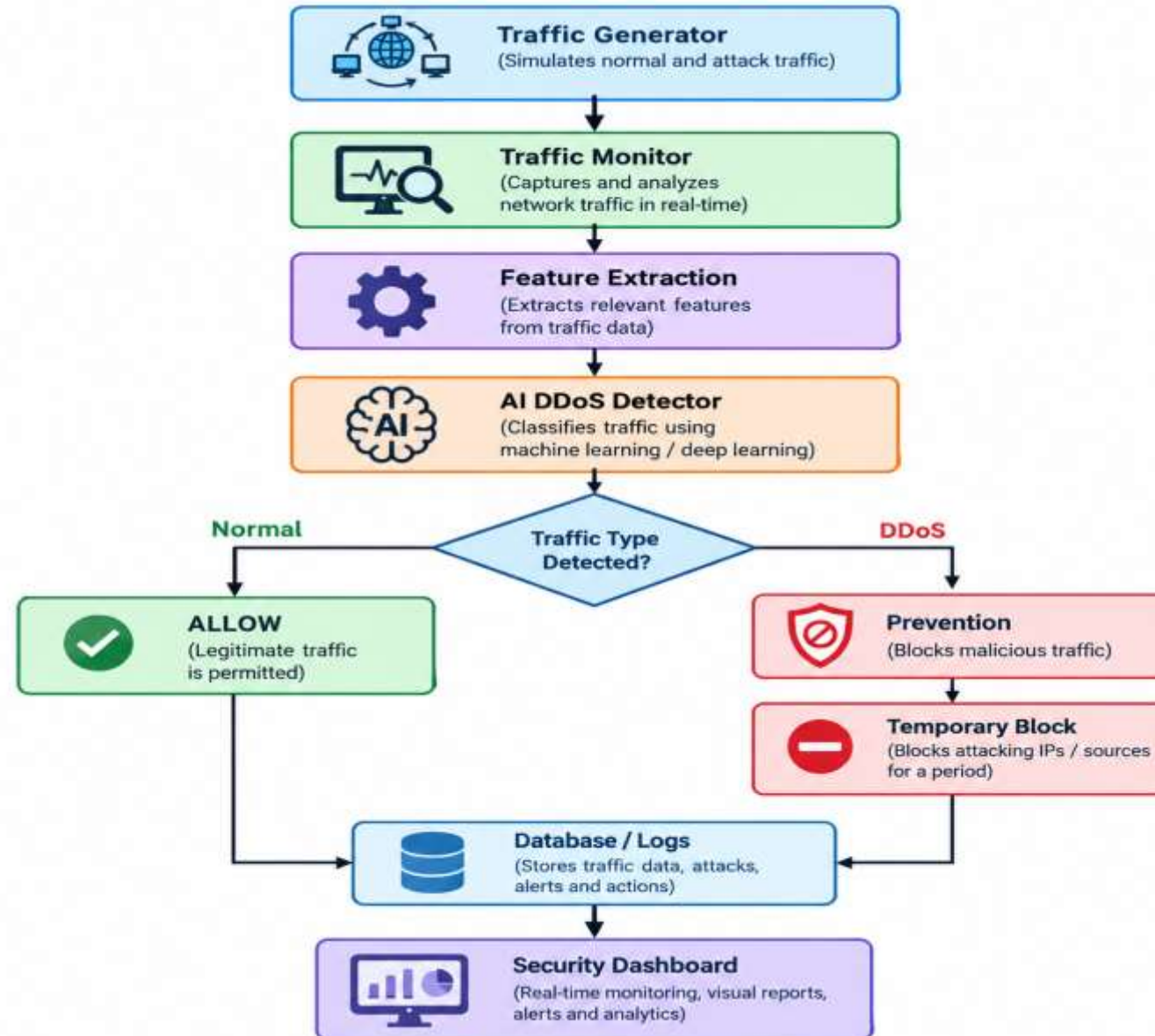
Methodology Flow



System Architecture Components

- **Traffic Generator** – produces controlled network traffic.
- **Traffic Monitor** – collects and analyzes traffic activity.
- **Feature Extractor** – converts traffic data into ML features.
- **AI Detector** – classifies traffic using the trained Random Forest model.
- **Prevention Engine** – automatically blocks detected malicious sources.

Architecture Flow



Module 1 — Network Traffic Simulation & Monitoring

- Creates a local software-based network environment for testing DDoS behavior.
- Generates both normal traffic and suspicious/high-volume traffic.
- Simulates multiple devices using different source IP addresses.
- Continuously monitors requests, packets, bytes, and traffic rates.
- Maintains a 5-second traffic analysis window for real-time monitoring.

Module 2 — AI-Based DDoS Detection

- Converts monitored network traffic into machine-learning features.
- Uses features such as packet rate, byte rate, connection/source rate, and request rate.
- Generates a labeled dataset containing Normal and DDoS traffic patterns.
- Uses a Random Forest Classifier for traffic classification.
- Trains the model using historical/simulated network traffic data.

Module 3 — Automatic DDoS Prevention & Security Response

- Receives the DDoS prediction from the AI detection engine.
- Identifies the source IP/device associated with suspicious traffic.
- Automatically applies a temporary blocking mechanism.
- Prevents further traffic from a blocked source during the mitigation period.

Module 4 — Security Operations Dashboard & Device Intelligence

- Provides a real-time Security Operations Center (SOC)-style dashboard.
- Displays traffic status, request rate, packet rate, AI confidence, and threat counts.
- Visualizes live network traffic analytics using a continuously updated chart.
- Displays simulated device locations and geographic network activity.
- Shows device IP, device ID, location, traffic rate, AI result, and security status.

Implementation



Automatic Prevention & Response

- The prevention module receives the ML prediction.
- When traffic is classified as DDoS, the source IP is identified.
- The malicious IP is added to a temporary block list.
- In our implementation, the default blocking period is 15 seconds.
- Requests from blocked sources are rejected during the blocking period.
- Expired blocks are automatically removed.

System Output & Performance

- Successfully generates normal and DDoS-like traffic locally.
- Provides real-time traffic monitoring through the dashboard.
- ML model performs automatic traffic classification.
- Suspicious source IPs are identified at device level.
- Automatic temporary blocking works without manual intervention

Limitations of the Proposed System

- The system uses simulated network traffic instead of live enterprise traffic.
- Detection accuracy depends on the quality and representativeness of the training dataset.
- The current implementation supports Normal and DDoS classification only.
- Temporary IP blocking is implemented for simulation purposes.
- The system is designed for localhost/software-only deployment.
- It does not currently integrate with real routers, firewalls, or IDS/IPS hardware.

Advantages & Disadvantages

Advantages

- Provides automated DDoS detection using Machine Learning.
- Reduces dependence on manual traffic monitoring.
- Enables automatic response to detected malicious sources.
- Provides real-time network visibility through the dashboard.
- Tracks source devices and simulated geographic locations.

Disadvantages

- Current implementation is primarily a controlled local simulation.
- Simulated locations do not represent real-world IP geolocation.
- Detection performance depends on the quality of training data.
- A trained model may not identify every new or unseen attack pattern.
- The current prevention mechanism is designed for the simulator.

Future Scope

- Integrate real network packet capture and live network interfaces.
- Support additional DDoS attack categories and traffic protocols.
- Use larger real-world network-security datasets for model training.
- Compare multiple Machine Learning and Deep Learning algorithms.
- Add real IP reputation and geolocation services.

Conclusion

- The project demonstrates an integrated approach to DDoS detection and prevention.
- Network traffic is continuously monitored and converted into useful features.
- Machine Learning is used to identify suspicious traffic patterns.
- The Random Forest model classifies traffic as Normal or DDoS.
- Detected malicious sources can be automatically blocked.

References

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THANK YOU...